PROBLEM 1.10

Laminar Viscous Poiseuille Flow from Navier-Stokes

Original statement · ProblemsCh1.pdf, p. 2

Navier-Stokes Reduction • Centerline Regularity • No-Slip Wall

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: In general, it is not possible to find explicit solutions of the Navier-Stokes equations that govern fluid flow. However, in certain cases the equations simplify very strongly. Suppose that a fluid is flowing down a pipe that has a circular cross-section of radius a. Assuming that the velocity V of the fluid depends only on its distance from the centre of the pipe, the equation satisfied by V is:

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = -P$$

where $P > 0$.

Multiply by r and integrate once to show that:

$$\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r}$$

where c is an arbitrary constant.

Integrate again to find an expression for the velocity, and then use the facts that:
(i) the velocity should be finite at all points in the pipe, and
(ii) fluids 'stick' to boundaries (which means that $V(a) = 0$),

to show that:

$$V(r) = \frac{P}{4}\left(a^2 - r^2\right)$$

(Poiseuille flow).

PHASE 2: FUNDAMENTAL FORMULATION

This is a linear second-order Boundary Value Problem (BVP) in radial coordinate $r \in [0, a]$. The parameter $P \equiv -\frac{1}{\mu}\frac{dp}{dz} > 0$ represents the pressure driving force per unit dynamic viscosity.

CLASSIFICATION WITH THE CLASS NOTES [NOTES, P. 2]

Expanding $\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = V'' + \frac{1}{r}V'$, the equation is $V'' + \frac{1}{r}V' = -P$: an ODE (one independent variable, r), of order 2, linear (form (L) with $a_0 = 1, a_1 = \frac{1}{r}, a_2 = 0, b = -P$) and inhomogeneous. Written as $\frac{d}{dr}\left(rV'\right) = -Pr$, it is solved, like the free fall of the notes, by integrating twice.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: First Integration (Velocity Gradient):

Multiply the differential equation by r:

$$\frac{d}{dr}\left(r\frac{dV}{dr}\right) = -Pr$$

Integrate directly with respect to r:

$$\int \frac{d}{dr}\left(r\frac{dV}{dr}\right)dr = \int -Pr\,dr \implies r\frac{dV}{dr} = -P\frac{r^2}{2} + c$$

Dividing by r ($r > 0$):

$$\boxed{\frac{\mathbf{dV}}{\mathbf{dr}} = -\frac{\mathbf{Pr}}{\mathbf{2}} + \frac{\mathbf{c}}{\mathbf{r}}} \qquad (1.10.1)$$

which rigorously proves the required intermediate formula.

Step 2: Second Integration (Velocity Field), keeping c:

Integrate (1.10.1) with respect to r on $r > 0$, using $\int r\,dr = \frac{r^2}{2}$ and $\int \frac{dr}{r} = \ln r$:

$$V(r) = \int \left(-\frac{Pr}{2} + \frac{c}{r} \right) dr = -\frac{P}{2} \cdot \frac{r^2}{2} + c \ln r + C_2 = -\frac{Pr^2}{4} + c \ln r + C_2 \quad \boxed{(1.10.2)}$$

where C_2 is the second constant of integration.

Step 3: Condition (i), Finite Velocity at the Centerline ($r = 0$):

The velocity must be finite at every point of the pipe, including $r = 0$. Since $\lim_{r \to 0^+} r^2 = 0$ but $\lim_{r \to 0^+} \ln r = -\infty$, the term $c \ln r$ makes $V(r) \to \mp\infty$ for any $c \neq 0$ (and $\frac{dV}{dr} = \frac{c}{r}$ diverges as well, so the shear stress $\tau_{rz} = \mu \frac{dV}{dr}$ would be unbounded). Finiteness therefore requires:

$$\mathbf{c = 0} \implies V(r) = -\frac{Pr^2}{4} + C_2 \quad \boxed{(1.10.3)}$$

Step 4: Enforcement of No-Slip Boundary Condition ($r = a$):

Condition (ii) states that viscous fluids adhere to solid surfaces without slip: $V(a) = 0$. Imposing this on equation (1.10.3):

$$V(a) = -\frac{Pa^2}{4} + C_2 = 0 \implies \mathbf{C_2 = \frac{Pa^2}{4}}$$

Substituting C_2 back into (1.10.3):

$$V(r) = \frac{Pa^2}{4} - \frac{Pr^2}{4} = \frac{\mathbf{P}}{\mathbf{4}} \left(\mathbf{a^2 - r^2} \right) \quad \boxed{(1.10.4)}$$

PHASE 4: FINAL RESULT & AEROSPACE QUANTITIES

BOXED RESULT

$$\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r} \implies V(r) = \frac{P}{4} \left(a^2 - r^2 \right)$$

Engineering Metrics:

- Centerline peak velocity: $V_{\max} = V(0) = \frac{Pa^2}{4}$.

- Volumetric Flow Rate (Hagen-Poiseuille Law), integrating over the annular elements $dA = 2\pi r\, dr$ with Barrow's rule:

$$Q = \int_0^a \frac{P}{4}\left(a^2 - r^2\right) 2\pi r\, dr = \frac{\pi P}{2}\int_0^a \left(a^2 r - r^3\right) dr = \frac{\pi P}{2}\left[\frac{a^2 r^2}{2} - \frac{r^4}{4}\right]_0^a = \frac{\pi P}{2}\left(\frac{a^4}{2} - \frac{a^4}{4}\right) = \frac{\pi P a^4}{8}$$

  With $P = -\frac{1}{\mu}\frac{dp}{dz}$ (see Phase 2) this gives $Q = \frac{\pi a^4}{8\mu}\left(-\frac{dp}{dz}\right)$.
- Wall shear stress: $\tau_w = -\mu \left.\frac{dV}{dr}\right|_{r=a} = \frac{\mu P a}{2} = \frac{a}{2}\left(-\frac{dp}{dz}\right)$.